Ex no:- 7

Implement Packet Sniffing using raw sockets in python.

Aim:-

* To implement a Packet sniffer using raw sockets in Python to capture and analyze network packets.

Requirements:

* Python 3.x
* windows/Linux OS * Python sockets.

PROCEDURE:

* Open cmd prompt with administrator.
* Create Packet-Sniffer_Py.
* Import the required socket.
* Obtain the local computer's IP address.
* Bind the raw socket to the local IP address.
* Enable IP header inclusion using setsocket
* Display the captured-Packet information

CODE:-

```
import socket
import struct
def Parse_ip_header (data)
```

```
ip_header= struct.unpack("!BBHHHBBH4s4s",
                         data[:20])
version_ihl= ip_header[0]
H = ip_header[5]
source_ip= socket.inet_ntoa(ip_header[8])
dest_ip= socket.inet_ntoa(ip_header[9])
return source_ip, dest_ip, H, protocol

S = socket.socket(socket.AF_INET, socket.
                  SOCK_RAW)

host name = socket.gethostname()
local_ip = socket.get host by name(hostname)

S.ioctl(socket.SIO_RCVALL, socket.RCVALL_ON)

by:
    while True:
        raw_data, addr = S.recvfrom(65535)
        source_ip, dest_ip, H,

    except Keyboard Interrupt:
        print("stopping sniffer")

finally:
    S.ioctl(socket.SIO_RCVALL, socket.
                        RCVALL_OFF)
    S.close().
```

RESULT:-

*Thus the packet sniffer was successfully implemented using raw sockets in Python.